



RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING LAB MANUAL**

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE
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INDEX

S.No.	Date	Title
1.	21/1/25	Azure Devops Environment Setup.
2.	21/1/25	Azure Devops Project Setup and User Story Management.
3.	28/1/25	Setting Up Epics, Features, And User Stories for Project Planning.
4.	11/2/25	Sprint Planning.
5.	18/2/25	Poker Estimation.
6.	25/2/25	Designing Class and Sequence Diagrams for Project Architecture.
7.	04/3/25	Designing Architectural and ER Diagrams for Project Structure.
8.	25/3/25	Testing – Test Plans and Test Cases.
9.	15/4/25	Load Testing and Pipelines.
10.	22/4/25	GitHub: Project Structure & Naming Conventions.

EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

Aim:

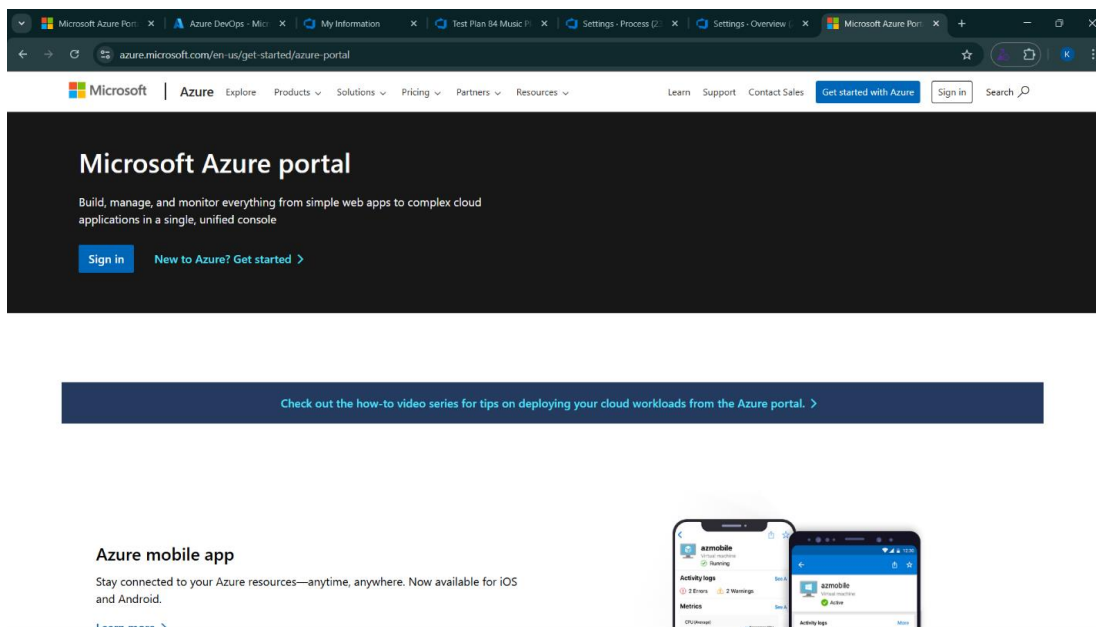
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

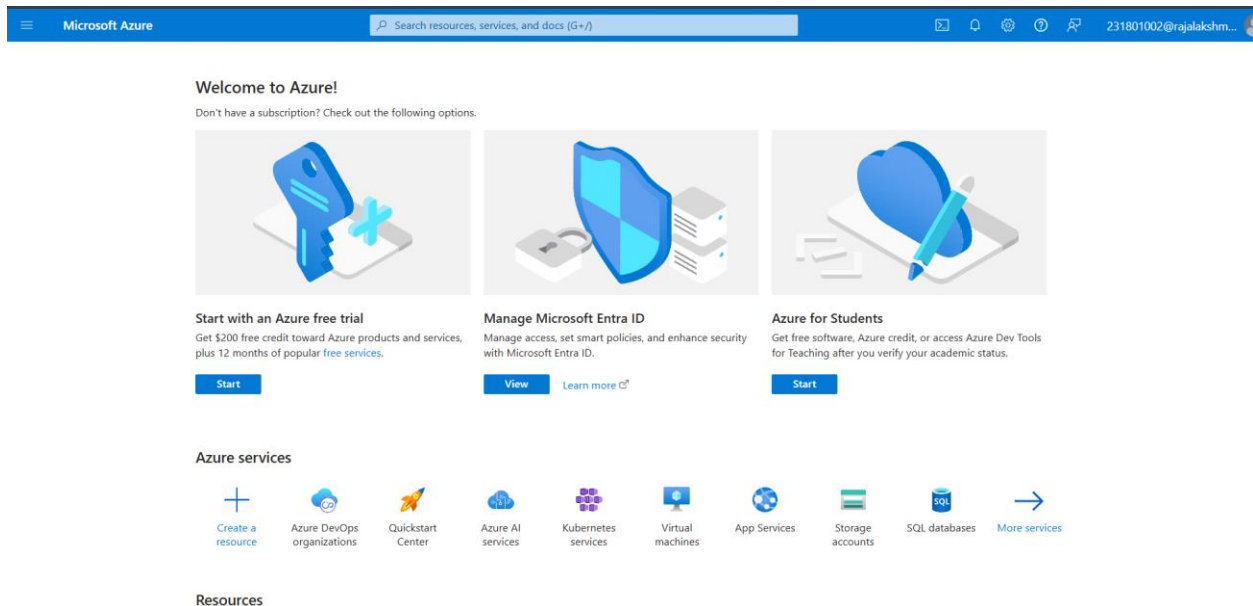
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

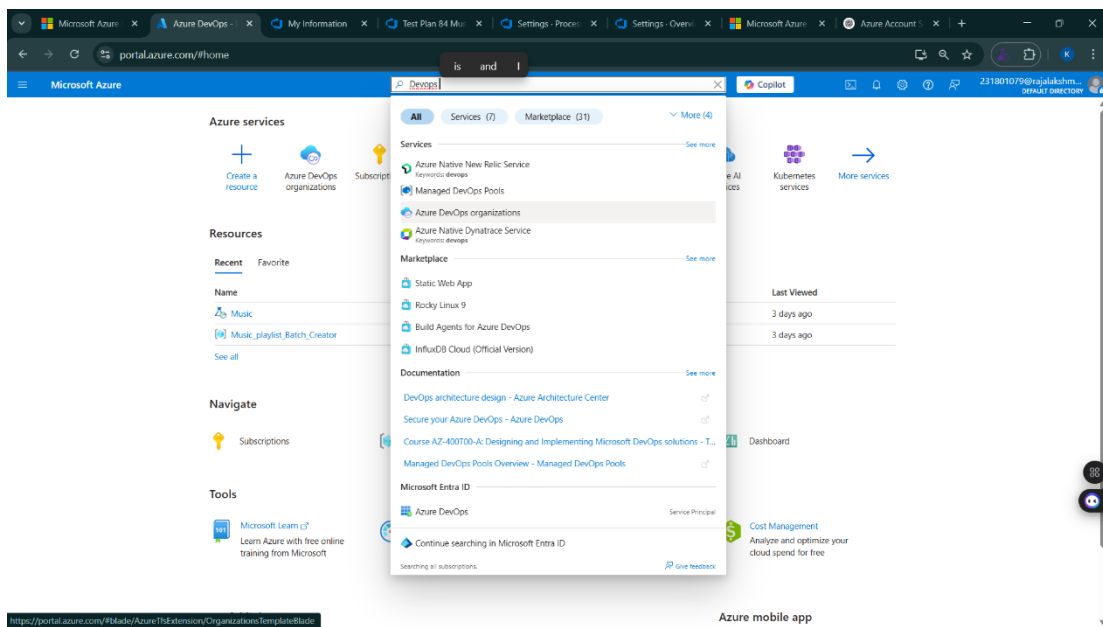
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



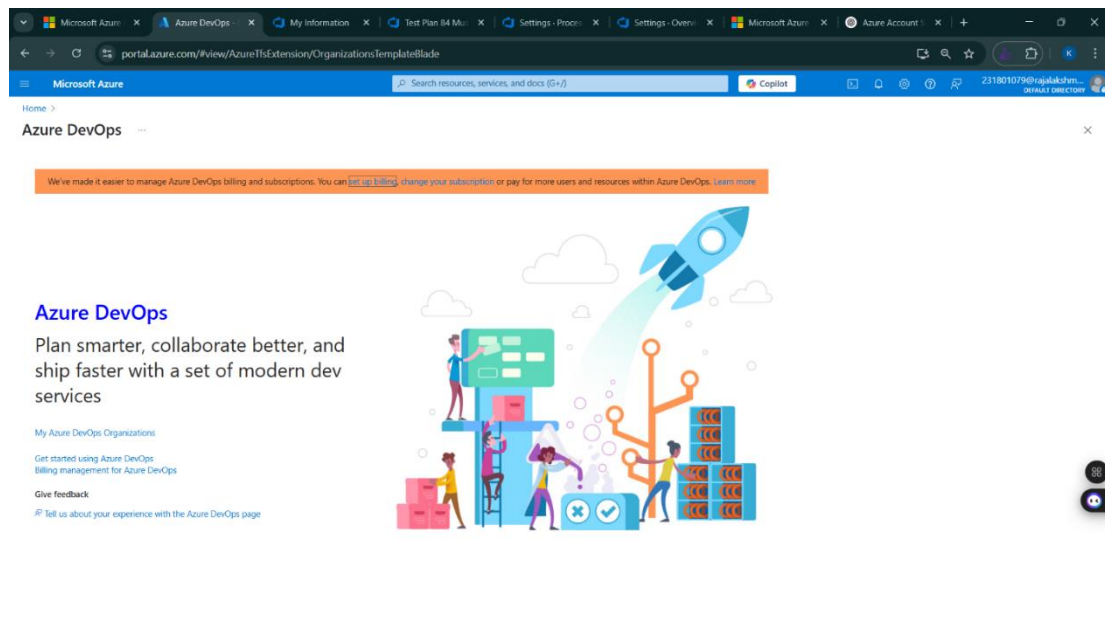
2. Azure home page



3. Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

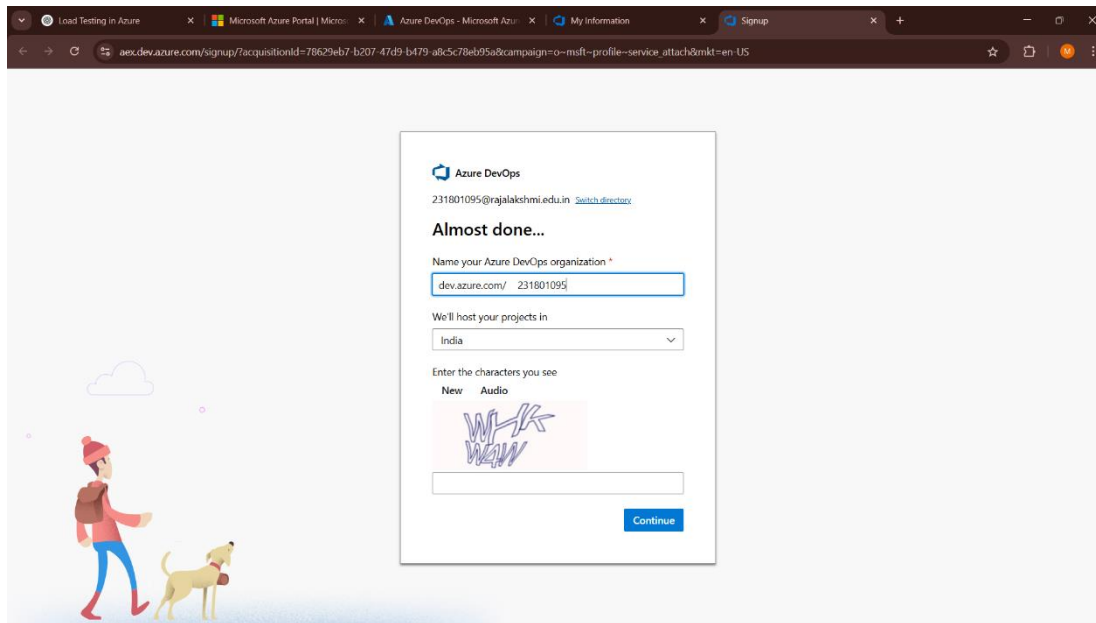
EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

Create new project

×

Project name *

Music Playlist Batch Creator

Description

Visibility



Public

Anyone on the internet can view the project. Certain features like TFVC are not supported.



Private

Only people you give access to will be able to view this project.

By creating this project, you agree to the Azure DevOps [code of conduct](#)

^ Advanced

Version control ?

Git

Work item process ?

231801095 Agile

Cancel

Create

3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Microsoft

Abhinav PrasadSign out

AP

Abhinav Prasad

231801002@rajalakshmi.edu.in

India

231801002@rajalakshmi.edu.in

Edit profile

Visual Studio Dev Essentials

Get everything you need to build and deploy your app on any platform.

Use your benefits

Azure DevOps Organizations

Create new organization

dev.azure.com/231801002 (Owner)

Projects

SMS-Text Messaging Service

SMS-Text 2

New project

Actions

Open in Visual Studio

dev.azure.com/231801032 (Member)

4.Project dashboard

231801002 / SMS-Text Messaging Service / Overview / Summary

Search

PublicInvite

SS

SMS-Text Messaging Service

About this project

Like 0

This project is a cloud-native SMS and text messaging application designed to provide seamless, secure, and real-time communication. Built with scalability and reliability in mind, the app enables users to send and receive messages, manage conversations, and connect through both one-on-one and group chats. It offers essential features such as user authentication, contact management, smart replies, and message scheduling, all supported by a robust cloud backend.

The system is engineered for performance and user experience, with real-time updates, secure data handling, and automated message processing. It targets both everyday users and small teams or businesses that rely on fast, dependable communication.

Project stats

Period: Last 7 days

Boards

0 Work items created

0 Work items

Members 6

APSS2SSBR

5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot shows the 'Backlog' view for the 'SMS-Text Messaging Service Team'. The interface includes a search bar at the top right, a 'New Work Item' button, and a 'View as Board' button. The main area displays a table of work items with columns for Order, Work Item Type, Title, State, Effort, Business Area, and Tags. The work items are organized into a hierarchy starting with an Epic 'SMS-Text Messaging Service', followed by Features like 'login page', 'Search feature', 'Interface', and 'Profile Page', each containing User Stories. The states of the work items are indicated by colored dots: New (blue), Closed (green), Resolved (blue), and Active (blue). A 'Planning' sidebar on the right shows a timeline for iterations, including 'iteration 428/03/2025 - 03/05/2025' and 'iteration 505/05/2025 - 10/06/2025', with planned effort and working days.

Order	Work Item Type	Title	State	Effort	Business Area	Tags
1	Epic	SMS-Text Messaging Service	New		Business	
	Feature	login page	Closed		Business	
	User Story	to provide the user a login page to log in	Resolved		Business	
	Feature	Search feature	Closed		Business	
	User Story	to provide searching feature for users	New		Business	
	Feature	Interface	Closed		Business	
	User Story	chatting interface for user to send messages to other ...	Active		Business	
	User Story	homepage interface	New		Business	
	Feature	Profile Page	New		Business	
	User Story	to change username and profile picture	New		Business	
	User Story	logout page	New		Business	
	Feature	grouping feature	New		Business	
	User Story	to provide the ability to make groups among users	New		Business	
	Bug	searched result is not visible	New		Business	

The screenshot shows a Microsoft account sign-in overlay. At the top, there is a search bar and navigation icons. The overlay displays the Microsoft logo, the user's name 'Abhinavu Prasad', and their email address '231801002@rajalakshmi.edu.in'. Below this, there is a blue circular profile picture with the initials 'AP' and a link to 'My Microsoft account'. A 'Sign in with a different account' button is visible at the bottom. The background shows a blurred view of the work item page.

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

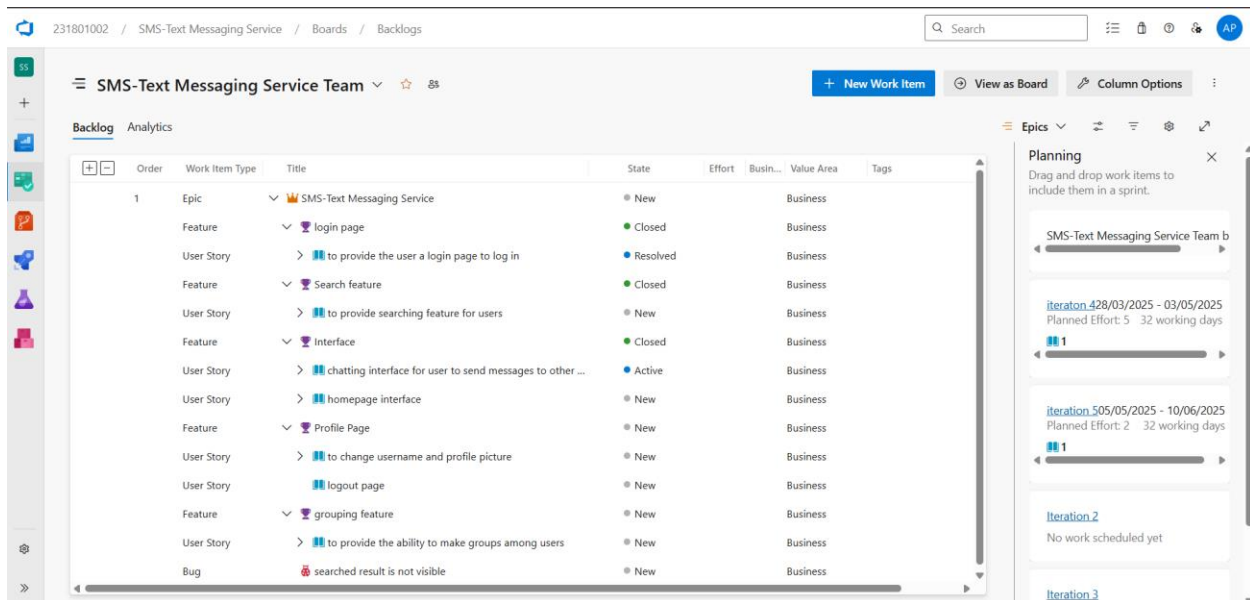
EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

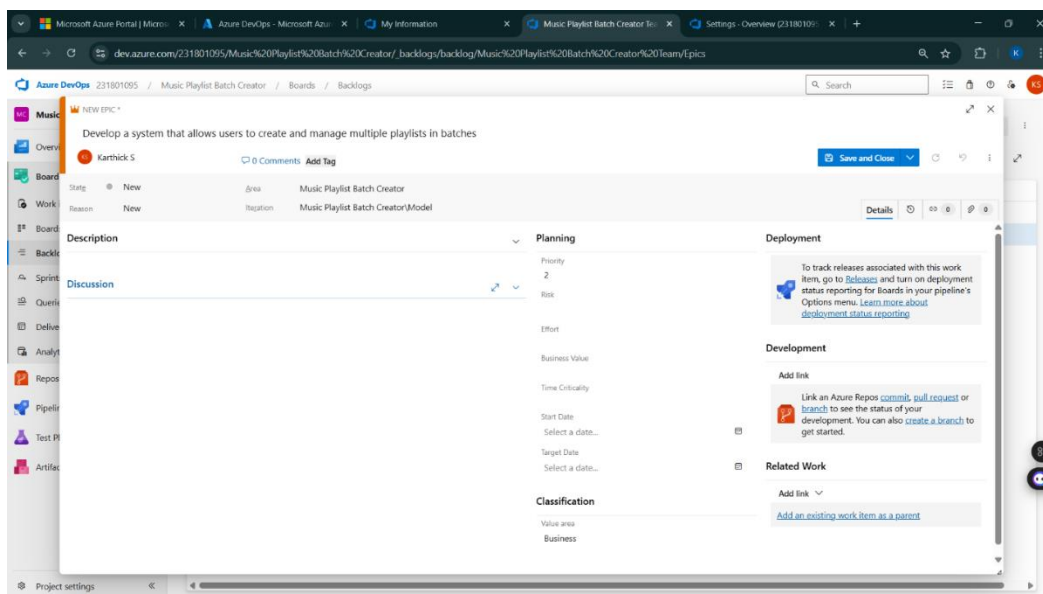


The screenshot shows the Azure DevOps interface for the 'SMS-Text Messaging Service Team'. The 'Backlog' view is active, displaying a list of work items. The table below represents the data shown in the backlog:

Order	Work Item Type	Title	State	Effort	Busin...	Value Area	Tags
1	Epic	SMS-Text Messaging Service	New			Business	
	Feature	login page	Closed			Business	
	User Story	to provide the user a login page to log in	Resolved			Business	
	Feature	Search feature	Closed			Business	
	User Story	to provide searching feature for users	New			Business	
	Feature	Interface	Closed			Business	
	User Story	chatting interface for user to send messages to other ...	Active			Business	
	User Story	homepage interface	New			Business	
	Feature	Profile Page	New			Business	
	User Story	to change username and profile picture	New			Business	
	User Story	logout page	New			Business	
	Feature	grouping feature	New			Business	
	User Story	to provide the ability to make groups among users	New			Business	
	Bug	searched result is not visible	New			Business	

On the right, the 'Planning' sidebar shows a timeline with iterations: 'iteration 428/03/2025 - 03/05/2025' (Planned Effort: 5, 32 working days) and 'iteration 505/05/2025 - 10/06/2025' (Planned Effort: 2, 32 working days). It also lists 'Iteration 2' with 'No work scheduled yet' and 'Iteration 3'.

1.Fill in Epics



The screenshot shows the 'NEW EPIC' form in Azure DevOps. The form is titled 'Develop a system that allows users to create and manage multiple playlists in batches'. The 'Description' field contains the text 'Develop a system that allows users to create and manage multiple playlists in batches'. The 'Planning' section shows 'Priority 2' and 'Risk'. The 'Deployment' section has a note: 'To track releases associated with this work item, go to Releases and turn on deployment status reporting for boards in your pipeline's Options menu. Learn more about deployment status reporting'. The 'Development' section has a note: 'Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.' The 'Related Work' section has a note: 'Add an existing work item as a parent'.

2.Fill in Features

FEATURE 2

2 login page

231801003

0 CommentsAdd Tag

Save and CloseFollow

Updated

StateClosedAreaSMS-Text Messaging ServiceReasonAcceptance tests passIterationSMS-Text Messaging ServiceDetails

Description

Discussion

AP

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Priority2RiskEffortBusiness ValueTime CriticalityStart DateSelect a date...Target Date

Deployment

To track releases assoc work item, go to [Relea](#) deployment status rep in your pipeline's Opti [more about deployme reporting](#)

Development

Add link

Link an Azure Repos [commit, pull request](#) or [branch](#) to s your development. You can also [create a branch](#) to get started

3.Fill in User Story Details

USER STORY 14

14 to provide searching feature for users

Dharshana Srinivasulu

0 CommentsAdd Tag

Save and CloseFollow

Updated by Dharshana Srinivasulu: Feb 26

Details30

Description

Click to add Description.

Acceptance Criteria

Click to add Acceptance Criteria.

Discussion

AP

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Planning

Story Points5Priority2Risk

Classification

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit, pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

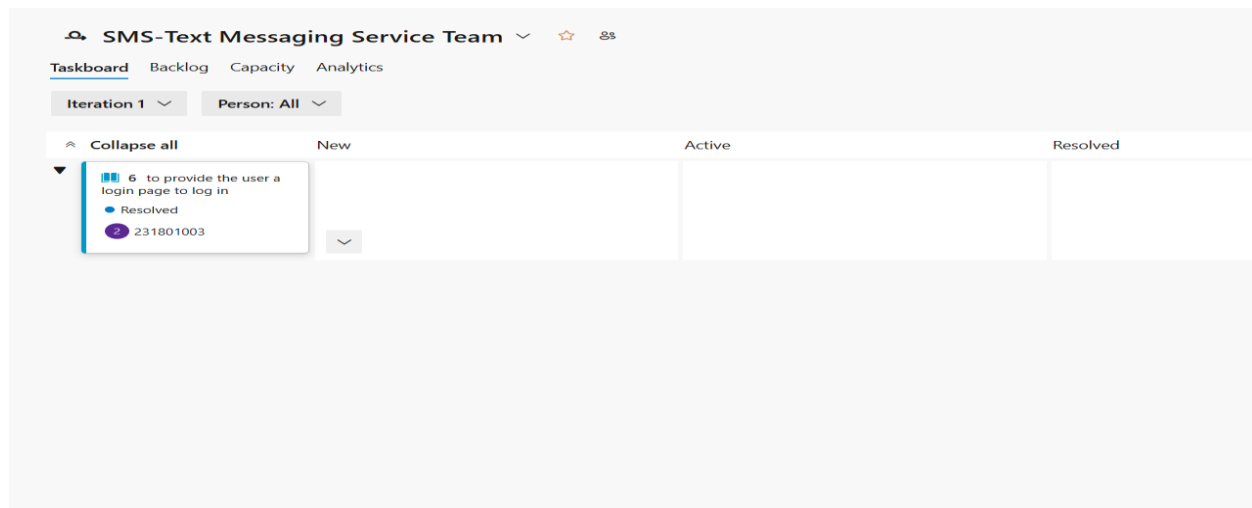
EXP NO: 4

SPRINT PLANNING

Aim:

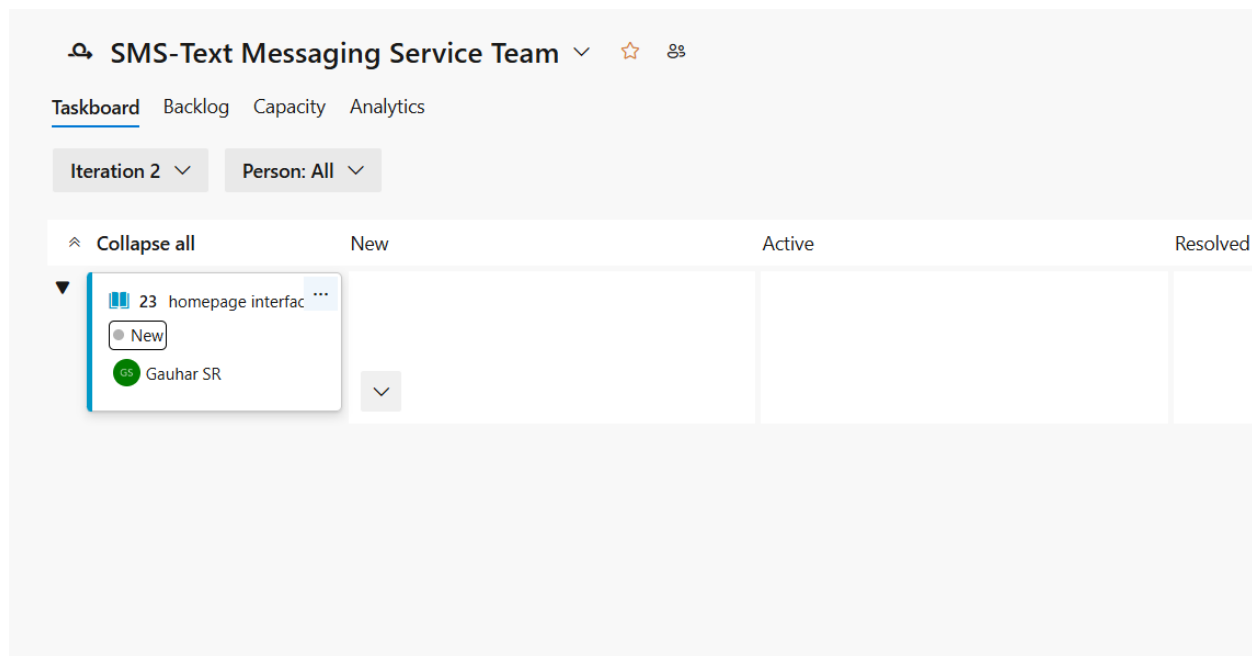
To assign user story to specific sprint for the Music Playlist Batch Creator Project.

Sprint Planning Sprint 1



The screenshot shows the Jira interface for the 'SMS-Text Messaging Service Team'. The 'Taskboard' tab is selected, and the view is set to 'Iteration 1' with 'Person: All'. The sprint board is divided into four columns: 'New', 'Active', and 'Resolved'. A card is visible in the 'New' column, titled '6 to provide the user a login page to log in', with a status of 'Resolved' and a priority of '231801003'. The card is expanded, showing a 'New' button and a 'Gauhar SR' label.

Sprint 2



The screenshot shows the Jira interface for the 'SMS-Text Messaging Service Team'. The 'Taskboard' tab is selected, and the view is set to 'Iteration 2' with 'Person: All'. The sprint board is divided into four columns: 'New', 'Active', and 'Resolved'. A card is visible in the 'New' column, titled '23 homepage interfac ...', with a status of 'New' and a priority of 'Gauhar SR'. The card is expanded, showing a 'New' button and a 'Gauhar SR' label.

Sprint 3

SMS-Text Messaging Service Team

Taskboard

Backlog

Capacity

Analytics

Iteration 3

Person: All

Collapse all

New

Active

Resolved

13 to change usernam and profile picture

New

Dharshana Srinivasulu

14 to provide searching feature for users

New

Dharshana Srinivasulu

Sprint 4

SMS-Text Messaging Service Team

Taskboard

Backlog

Capacity

Analytics

iteration 4

Person: All

New Work Item

Column Options

28 March - 3 May

5 work days remaining

Collapse all

New

Active

Resolved

Closed

7 chatting interface fo user to send messages to other user

Active

Boopesh R

16 to provide the ability to make groups among users

New

Abhinavu Prasad

2116231801002

CS23432

Result:

The Sprints are created for the Music Playlist Batch Creator Project.

EXP NO: 5

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - Music Playlist Batch Creator Project.

Poker Estimation

USER STORY 7

7 chatting interface for user to send messages to other user

Boopesh R

0 Comments Add Tag

Save and Close

Follow

Updated by Abhinav Prasad: Apr

State: Active

Area: SMS-Text Messaging Service

Reason: Acceptance tests fail

Iteration: SMS-Text Messaging Service\iteration 4

Details

2

0

Description

Click to add Description.

Acceptance Criteria

Click to add Acceptance Criteria.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Planning

Story Points
5

Priority
2

Risk

Classification

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Result:

The Estimation/Story Points is created for the project using Poker Estimation.

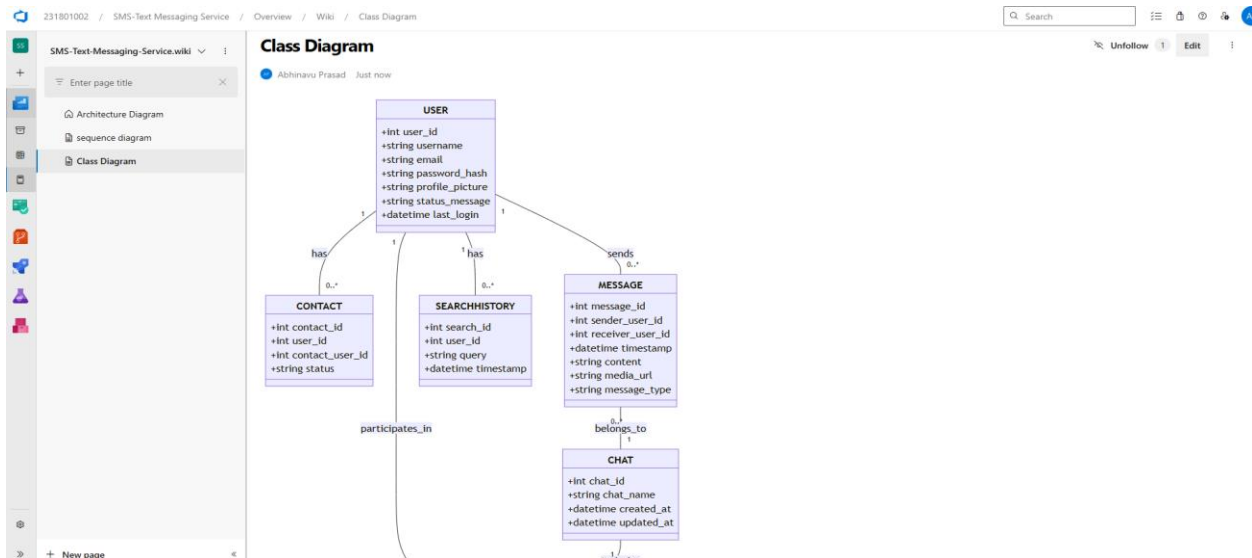
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

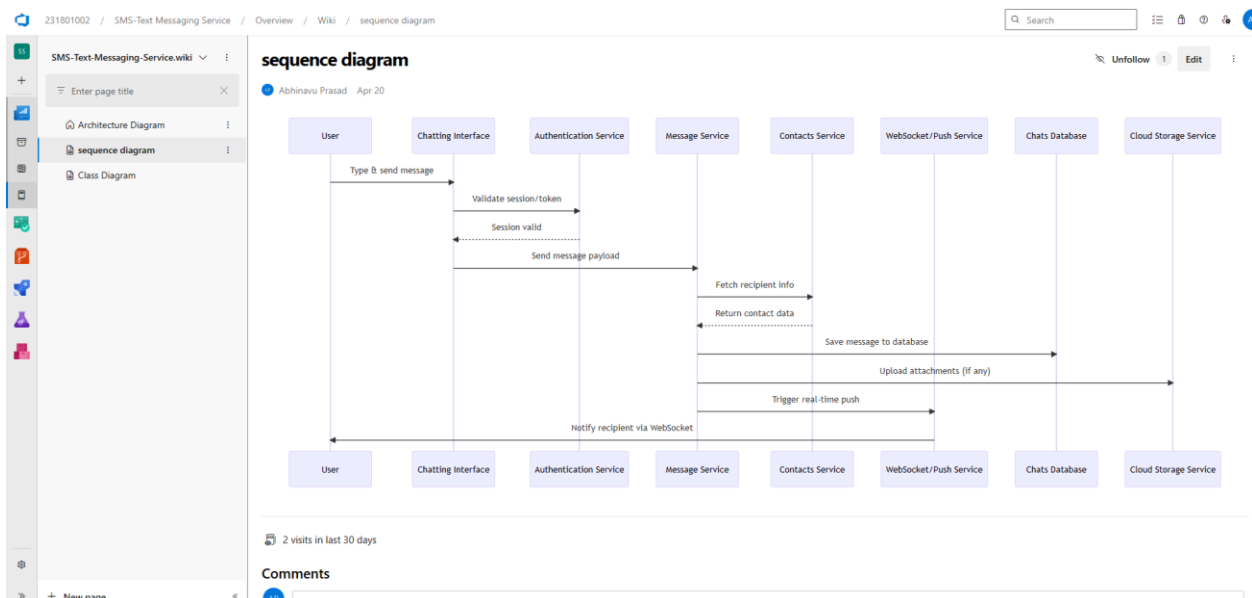
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Music Playlist Batch Creator.

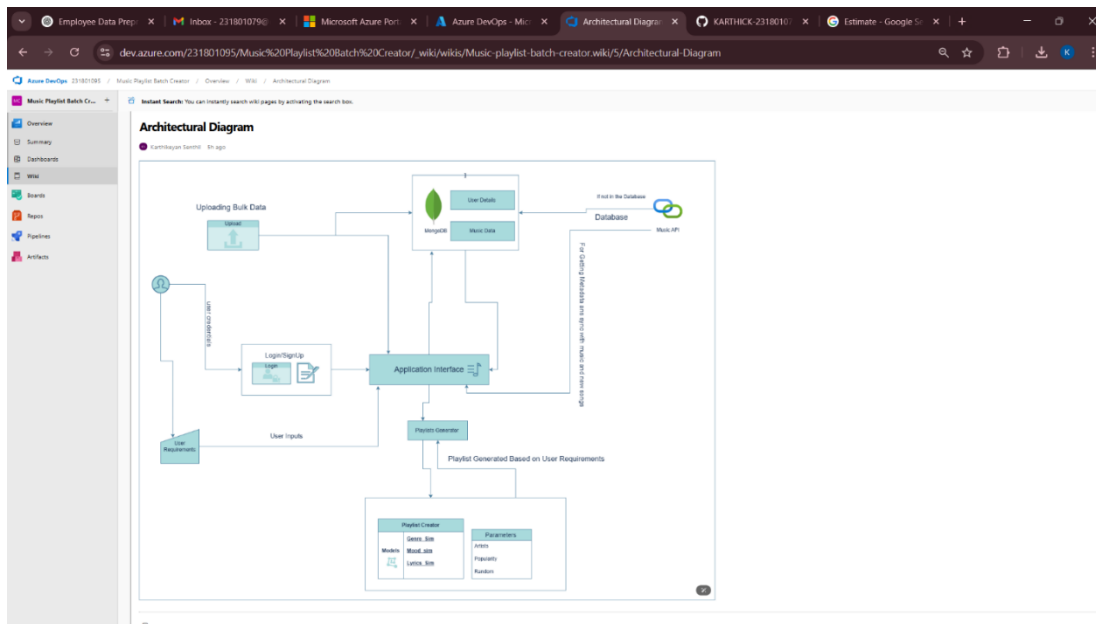
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

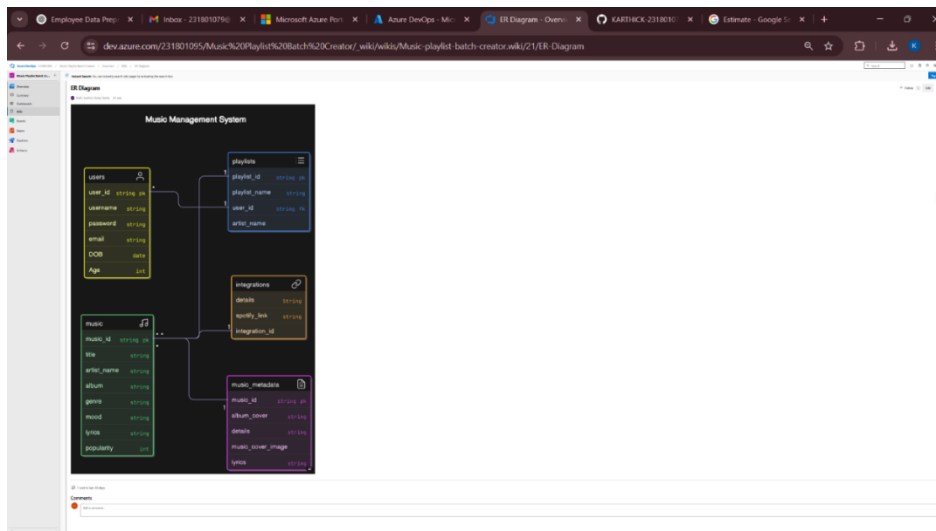
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Music Playlist Batch Creator

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
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Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Viewing and Managing Playlists
- Fetching Real-time Metadata
- Editing playlists (rename, reorder, record)
- Creating smart audio playlists based on categories (mood, genre, artist, etc.)

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly (e.g., TC01 – Successful Login, TC10 – Save Playlist Fails).
- Helps in quick identification and linking to user stories or features.

7. Separate Test Suites

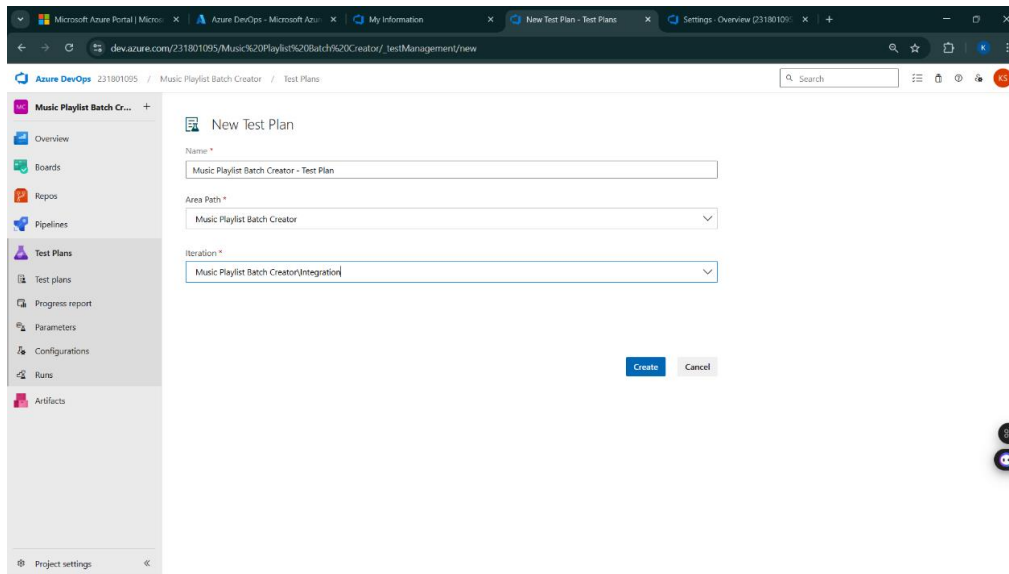
- Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).

- Improves organization and test execution flow in Azure DevOps.

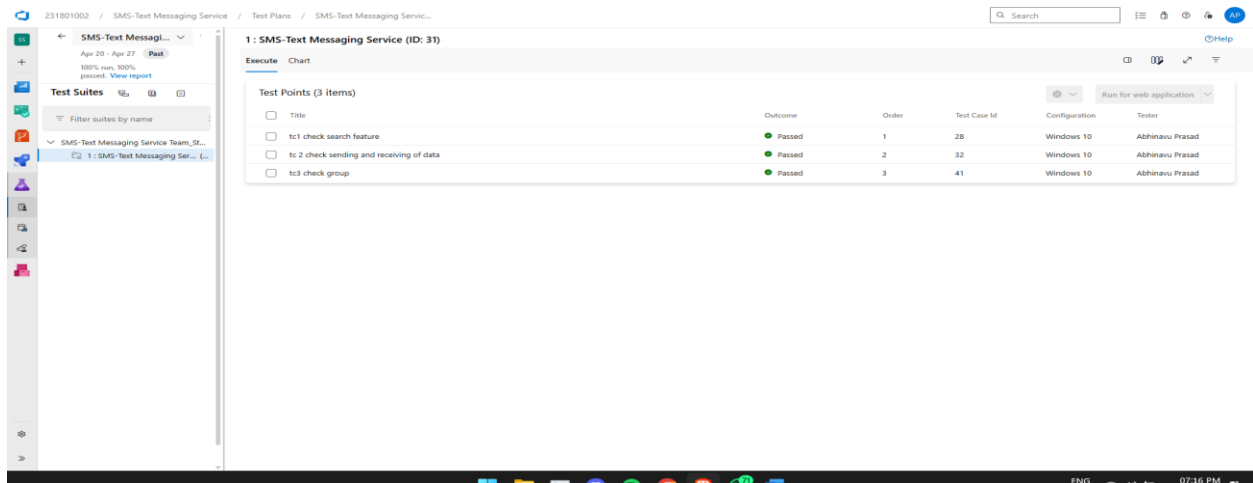
8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Batch Creator – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access my playlists (ID: 79).
- As a user, I need to see my playlist in one place (ID: 76).
- As a user, I should be able to create an audio playlist as needed (ID: 73).
- As a user, I should be able to rename, record, and change the playlist (ID: 68).
- As a user, I need to have real-time metadata (ID: 65).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. TC01 – Successful Sign Up

- **Action:**
 - Go to the Sign-Up page.
 - Enter valid name, email, and password.
 - Click "Sign Up".
- **Expected Results:**
 - Sign-Up form is displayed.
 - Fields accept values without error.
 - Account is created, and the user is redirected to the dashboard.
- **Type:** Happy Path

2. TC02 – Secure Login

- **Action:**
 - Go to the Login page.
 - Enter valid email and password.
 - Click on "Login".
- **Expected Results:**
 - Login form is displayed.
 - Fields accept data without error.
 - User is logged in and redirected to the dashboard.
- **Type:** Happy Path

3. TC03 – Sign Up with Existing Email

- **Action:**
 - Go to the Sign-Up page.
 - Enter a name and an already registered email.
 - Click on "Sign Up".
- **Expected Results:**

- Fields accept data.
- Error message "Email already registered" is displayed.
- **Type:** Error Path

4. TC04 – Login with Wrong Password

- **Action:**
 - Go to the Login page.
 - Enter valid email and incorrect password.
 - Click on "Login".
- **Expected Results:**
 - Input is accepted.
 - Error message "Invalid username or password" is shown.
- **Type:** Error Path

Test Suit: TS02 - View Playlists (ID: 87)

1. TC05 – View Playlist Page

- **Action:**
 - Log in successfully.
 - Navigate to "My Playlists" section.
- **Expected Results:**
 - All created playlists are displayed clearly.
- **Type:** Happy Path

2. TC06 – Playlist Loading Failure

- **Action:**
 - Disconnect from the internet.
 - Navigate to "My Playlists".
- **Expected Results:**
 - Network is offline.
 - Error message "Unable to load playlists" is shown.
- **Type:** Error Path

Test Suit: TS03 - Real-Time Metadata (ID: 88)

1. TC07 – Real-Time Metadata Display

- **Action:**
 - Play a song.
 - Observe the metadata panel.
- **Expected Results:**
 - Metadata (title, artist, album, duration) is displayed and updates in real time.
- **Type:** Happy Path

2. TC08 – Metadata Not Updating

- **Action:**

- Play a different song.
 - Observe the metadata panel.
- **Expected Results:**
 - Metadata remains static or shows default/fallback message.
- **Type:** Error Path

Test Suit: TS04 - Playlist Editing (ID: 89)

1. TC09 – Rename Playlist Successfully

- **Action:**
 - Navigate to "My Playlists".
 - Click "Rename" next to a playlist.
 - Enter a new name and click "Save".
- **Expected Results:**
 - Playlist name updates successfully.
- **Type:** Happy Path

2. TC10 – Rename with Blank Name

- **Action:**
 - Click "Rename" on a playlist.
 - Leave the field blank.
 - Click "Save".
- **Expected Results:**
 - Error message "Playlist name cannot be empty" is shown.
- **Type:** Error Path

3. TC11 – Change Playlist Order

- **Action:**
 - Open a playlist.
 - Drag and drop songs to reorder.
 - Click "Save".
- **Expected Results:**
 - Playlist order is updated and saved.
- **Type:** Happy Path

4. TC12 – Change Playlist Order Fails

- **Action:**
 - Login and go to "My Playlists".
 - Select a playlist.
 - Go offline or simulate server error.
 - Reorder songs and click "Save Order".
- **Expected Results:**
 - Error message: "Failed to update order. Please check your connection".
- **Type:** Error Path

Test Suit: TS05 - Smart Playlist Creation (ID: 90)

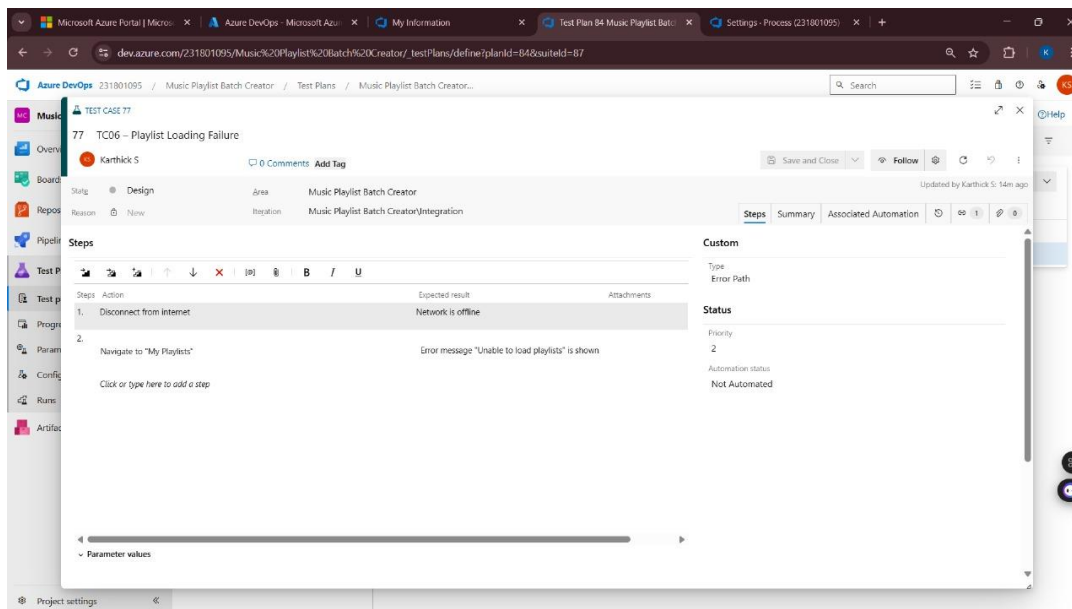
1. TC13 – Generate Playlist Based on Various Categories

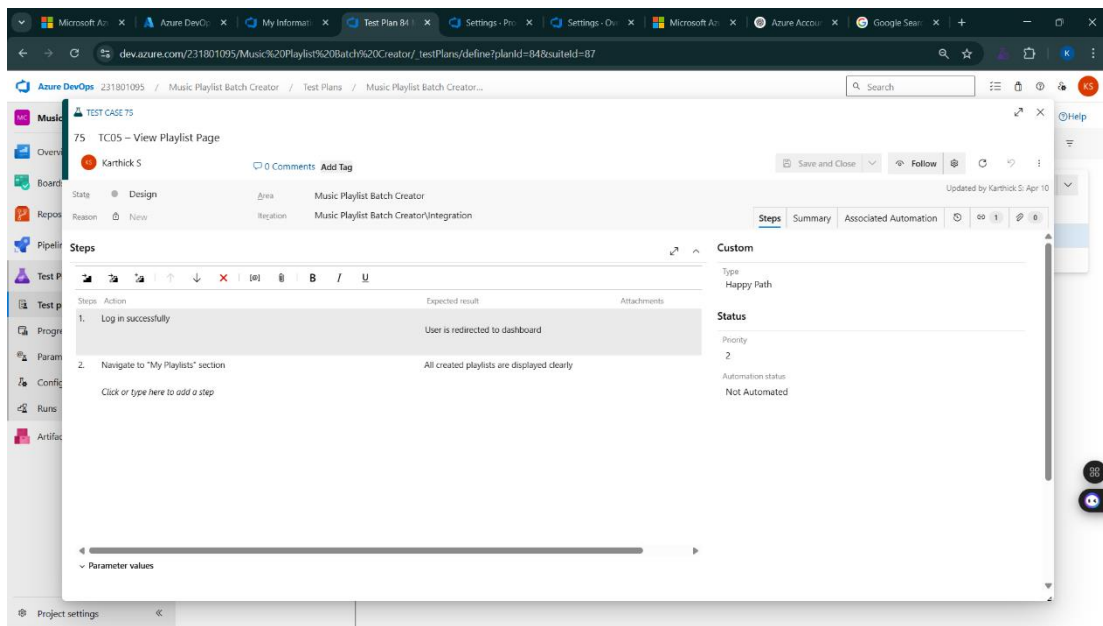
- **Action:**
 - Login with valid credentials.
 - Click on "Generate Playlist".
 - Select categories.
 - Click "Generate Playlist".
- **Expected Results:**
 - Playlist is generated based on selected mood and categories.
- **Type:** Happy Path

2. TC14 – Fail to Generate Playlist Due to Missing Category Selection or Invalid Input

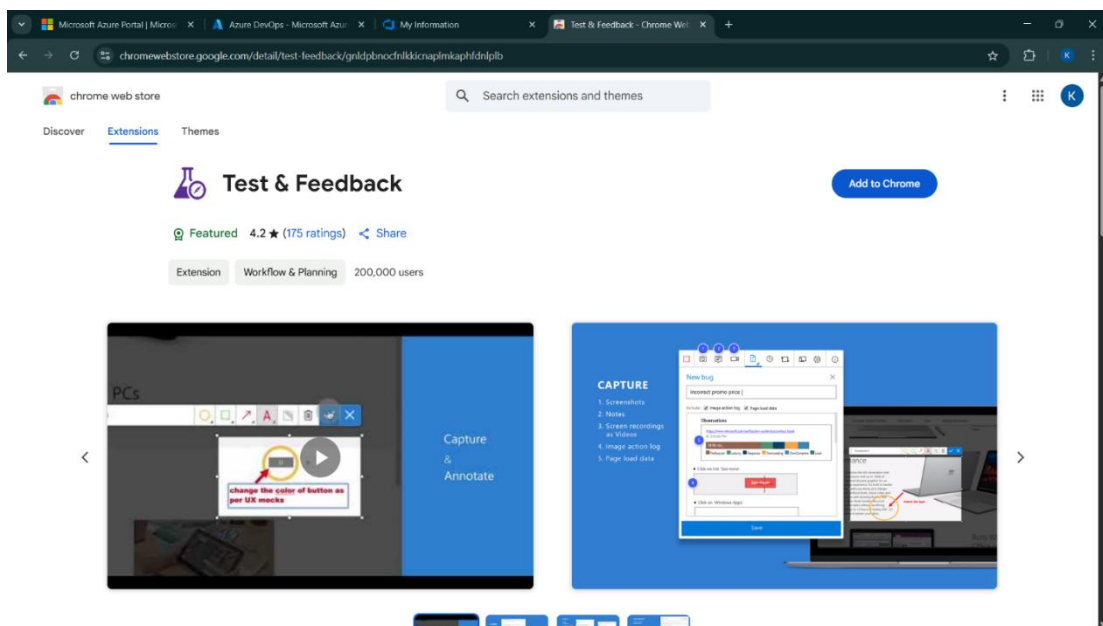
- **Action:**
 - Login with valid credentials.
 - Click on "Generate Playlist".
 - Select categories.
 - Click "Generate Playlist".
- **Expected Results:**
 - Error message: "Please select at least one valid category" or "No recommendations found for the selected filters".
- **Type:** Error Path

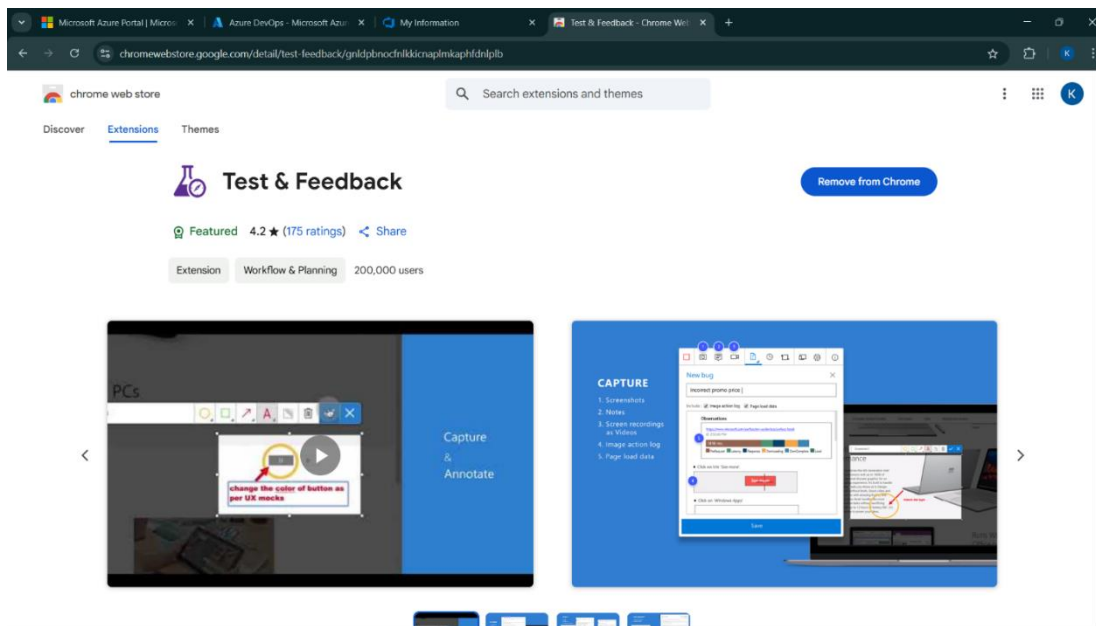
Test Cases



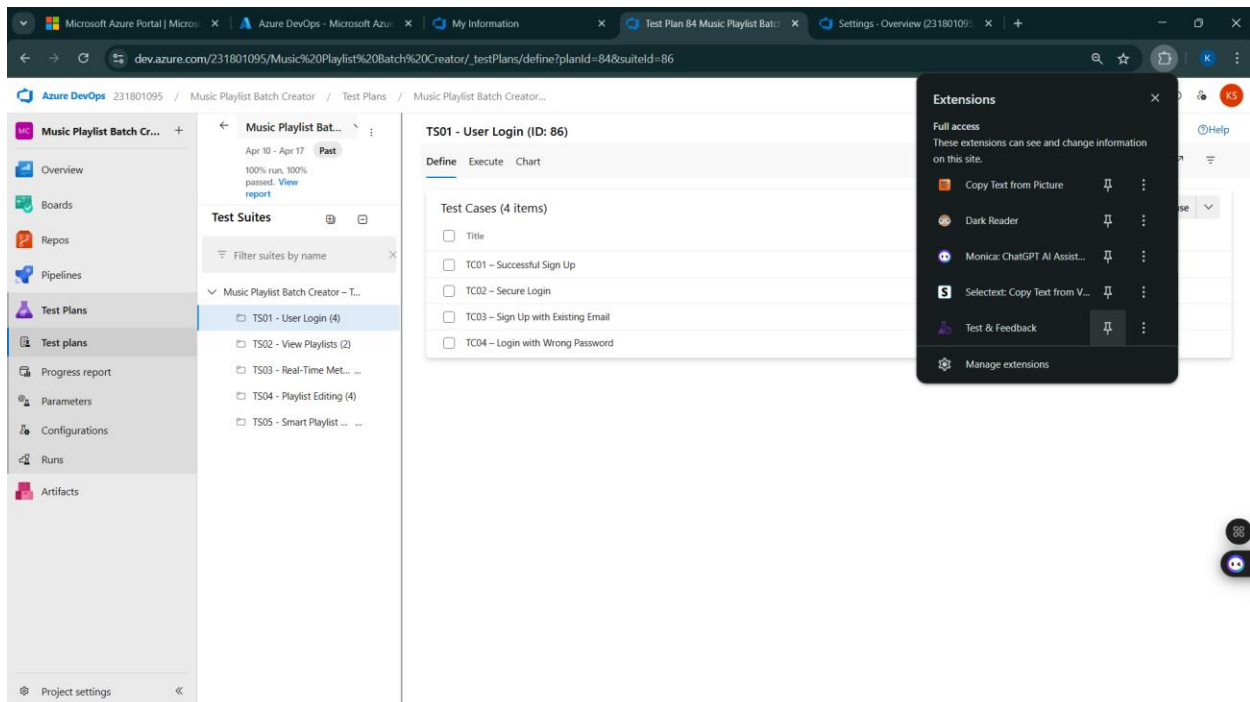


4.Installation of test





Test and feedback
Showing it as an extension



5. Running the test cases

Microsoft Azure Portal | M | Azure DevOps - Microsoft | My Information | Test Plan 84 Music Playlist | Settings - Overview (2318) | Bug Report Playlist Page

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testPlans/execute?planId=84&suiteId=87

Azure DevOps 231801095 / Music Playlist Batch Creator / Test Plans / Music Playlist Batch Creator...

Music Playlist Batch Cr... +

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Parameters

Configurations

Runs

Artifacts

Project settings

Music Playlist Bat... Apr 10 - Apr 17 Past 100% run, 100% passed. View report

Test Suites

Filter suites by name

Music Playlist Batch Creator - T...

TS01 - User Login (4)

TS02 - View Playlists (2)

TS03 - Real-Time Met...

TS04 - Playlist Editing (4)

TS05 - Smart Playlist ...

TS02 - View Playlists (ID: 87)

Define Execute Chart

Test Points (2 items)

Title	Outcome	Order	Test Case Id	Configuration	Tester
TC05 - View Playlist Page	Passed	1	75	Windows 10	Malu karthick B...
TC06 - Playlist Loading Failure	Passed	2	77	Windows 10	Malu karthick B...

View execution history

Mark Outcome

Run

Reset test to active

Edit test case

Assign tester

View test result

Run for web application

Run for desktop application

Run with options

Runner - Test Plans - Google Chrome

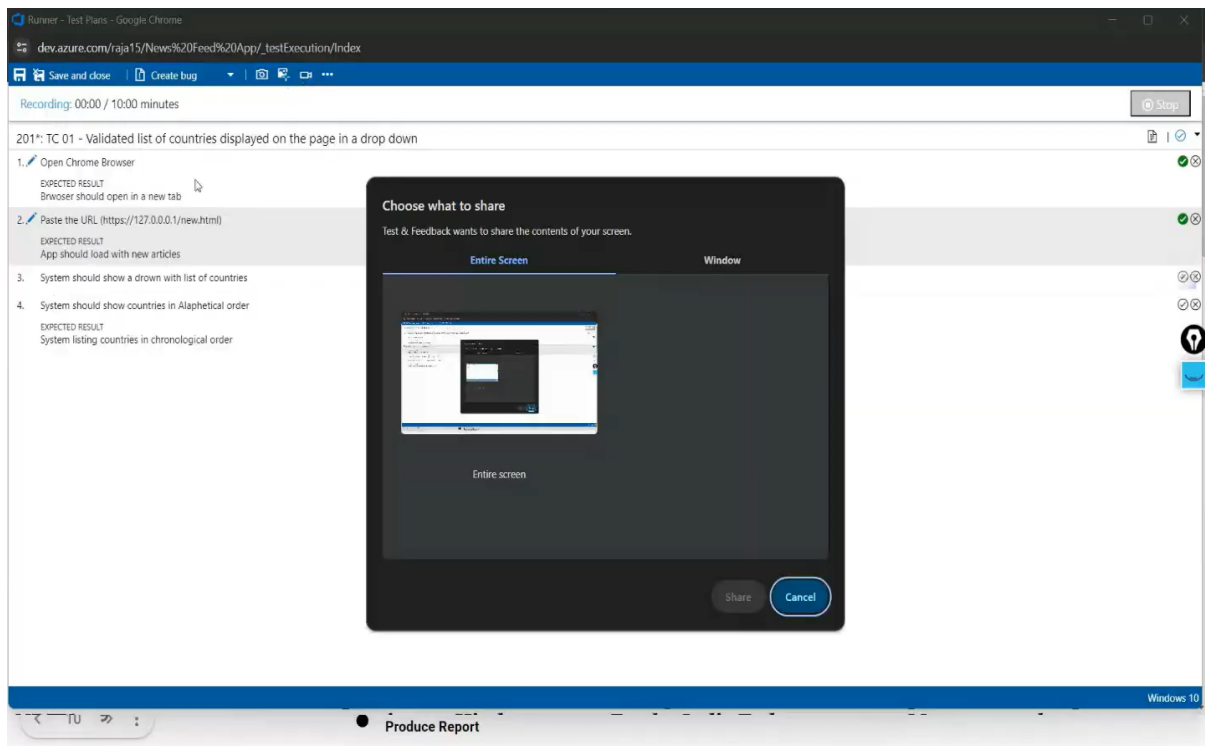
dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testExecution/Index

Save and close | Create bug

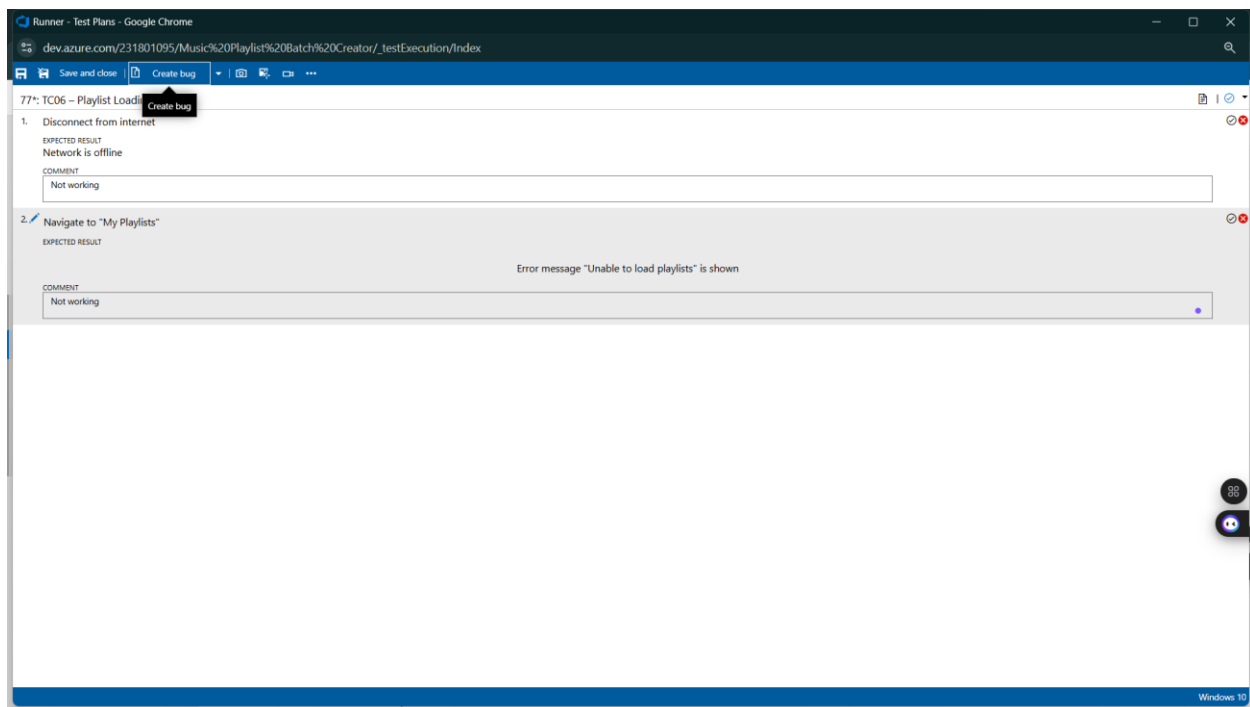
75: TC05 - View Playlist Page

- Log in successfully
EXPECTED RESULT
User is redirected to dashboard
- Navigate to "My Playlists" section
EXPECTED RESULT
All created playlists are displayed clearly

6. Recording the test case



7. Creating the bug



Runner - Test Plans - Google Chrome

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testExecution/Index

77: TC06 - Playlist Loading Failure

1. Disconnect from internet

NEW BUG*

TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Unassigned 0 comments Add tag TB01 - Playlist loading spinner keeps spinning indefinitely on poor network Save & Close

2. Navigation

State: New Reason: New Area: Music Playlist Batch Creator Iteration: Music Playlist Batch Creator

Repro Steps

18-04-2025 03:23 Bug filed on "TC06 - Playlist Loading Failure"

Step no.	Result	Title
1.	Failed	Disconnect from internet
		Expected Result
		Network is offline
2.	Failed	Comments: Page Not loading
		Navigate to "My Playlists"
		Expected Result

Error message "Unable to load playlists" is shown

Test Configuration: Windows 10

Planning

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

+ Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

+ Add link

Add an existing work item as a parent

Tested By

77 TC06 - Playlist Loading Failure

Updated 10-04-2025, @ Design

System Info

Found in Build

Microsoft Azure Portal x Azure DevOps - Mic x My Information x Test Plan 84 Music P x Runs - Test Plans x Settings - Overview x Bug Report Playlist x

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testManagement/runs?_a=resultSummary&runId=48&resultId=100000

Azure DevOps 231801095 / Music Playlist Batch Creator / Test Plans / Runs

Music Playlist Batch Cr... Enter Run ID... Go Run 48 - TS02 - View Playlists (Manual) / TC06 - Playlist Loading Failure

92 TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Bug 32 Unassigned 0 comments Add tag Save & Close Follow Details

Updated by Karthick S: 8m ago

System Info

Found in Build

Integrated in Build

Browser - Name	Google Chrome 135
Browser - Language	en-IN
Browser - Height	864
Browser - Width	1536
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x86_64
Operating system - Processor model	11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
Operating system - Number of processors	4
Memory - Available	814784512
Memory - Capacity	8216240128
Display - Pixels per inch (X axis)	120
Display - Pixels per inch (Y axis)	120
Display - Device pixel ratio	1.25

Discussion

Add a comment. Use # to link a work item, ! to link a pull request, or @ to mention a person.

Project settings

Name

SystemInformation-2025-04-18T03-23-58.166Z.json

Size

1K

8. Test case results

The screenshot shows the Azure DevOps interface for a test plan. The left sidebar contains navigation options like Overview, Boards, Repos, Pipelines, Test Plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The main area displays 'Test Suites' and 'Test Points'. A modal window titled 'TC05 - View Playlist Page' is open, showing 'Test Case Results'.

Outcome	TimeStamp	Configuration	Run by	Tester	Test P
Passed	4m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Not Applicable	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	14m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Tuesday	Windows 10	Karthikayan Senthil	Mallu karthick Balaji ...	Music
Passed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Failed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music

9. Test report summary

The screenshot shows the Azure DevOps Work Items page for a bug report. The bug is titled 'BG 01 - Countries Drop down Not Available on the page' and is assigned to 'rajesh prabhu'. The state is 'New'. The repro steps are listed, and the results show that the first two steps passed, but the third step failed.

Repro Step

Step no.	Result	Title
1.	Passed	Open Chrome Browser
2.	Passed	Paste the URL (https://127.0.0.1/new.html)
3.	Failed	App should load with new articles

Expected Result

Browser should open in a new tab

Actual Result

App should load with new articles

Comments

Validated list of countries displayed on the

Planning

Resolved Reason

Story Points

Priority

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

- Assigning bug to the developer and changing state

Bug 92 TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Assigned to: Karthick S

Repro Steps:

- Failed** 18-04-2025 03:23 Bug filed on "TC06 - Playlist Loading Failure"
 - Step no. 1. **Failed** Disconnect from internet
 - Expected Result: Network is offline
 - Comments: Page Not loading
- Failed** 2. **Failed** Navigate to "My Playlists"
 - Expected Result: Error message "Unable to load playlists" is shown

Test Configuration: Windows 10

10.Progress report

Progress report

Music Playlist Batch Creator - Test Plan (+1)

Summary:

- 2 Test plans
- 28 Test points
- 28 (28 / 28) Test points run
- 100% Run
- 100% (28 / 28) Pass rate
- 28 Passed

Outcome trend (Last 14 Days):

Legend: Not run (Grey), Passed (Green)

Details:

Test plan name	Test points	Run %	Passed %	Failed %	Not run count
> Music Playlist Batch Creator Team_Stories_Integration	14	100	100	0	0
> Music Playlist Batch Creator - Test Plan	14	100	100	0	0

11.Changing the test template

Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan 84 Music Playlist x Settings - Process (231801 x Settings - Overview (231801 x +

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of process.	0

Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan 84 Music Playlist x Settings - Process (231801 x Settings - Overview (231801 x +

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

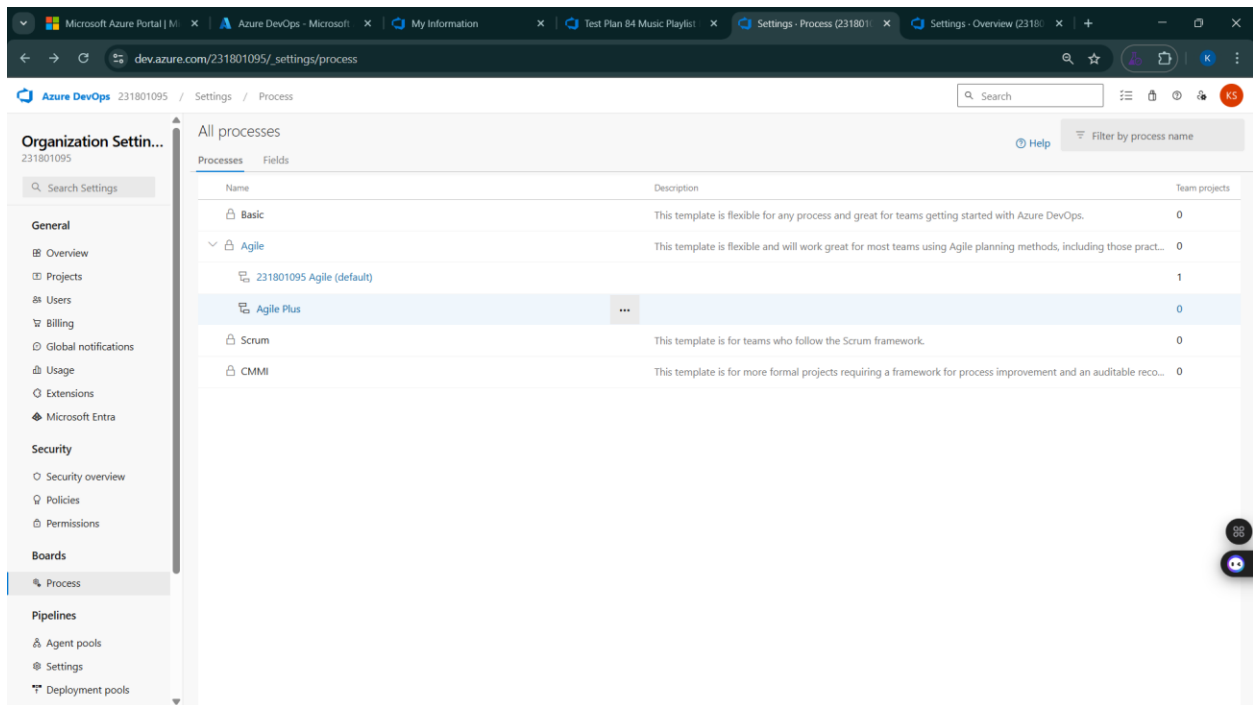
Pipelines

- Agent pools
- Settings
- Deployment pools

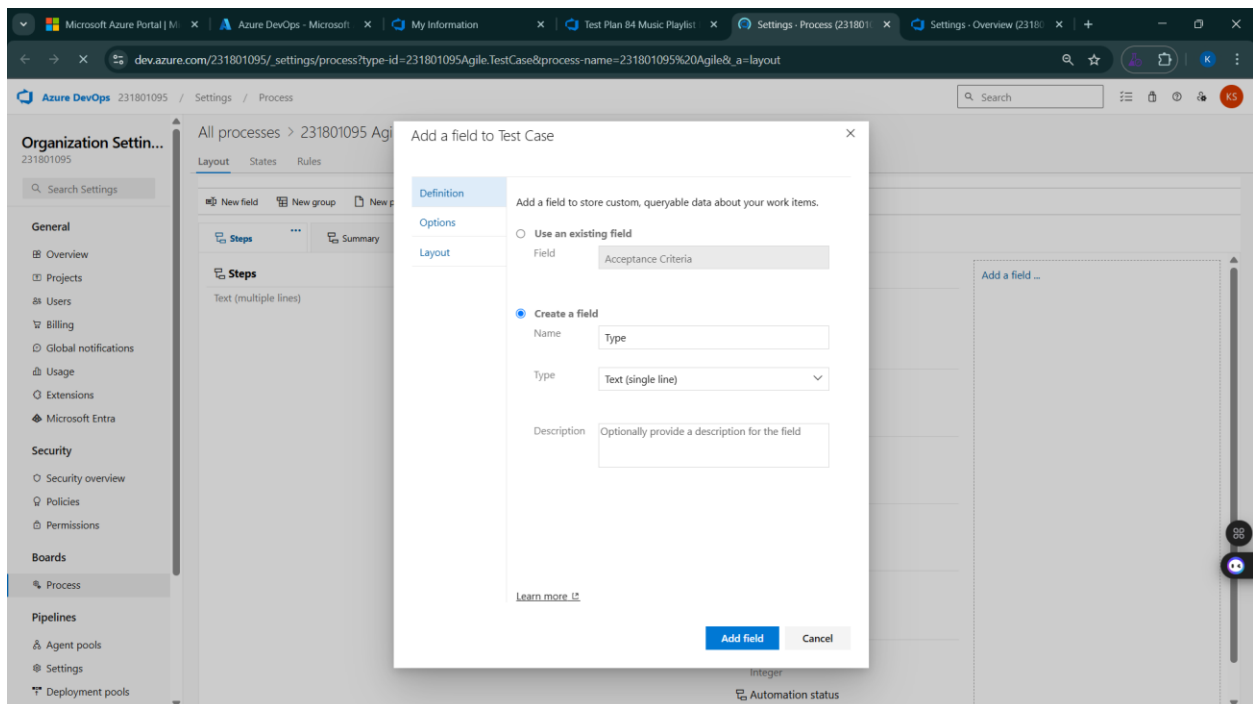
All processes

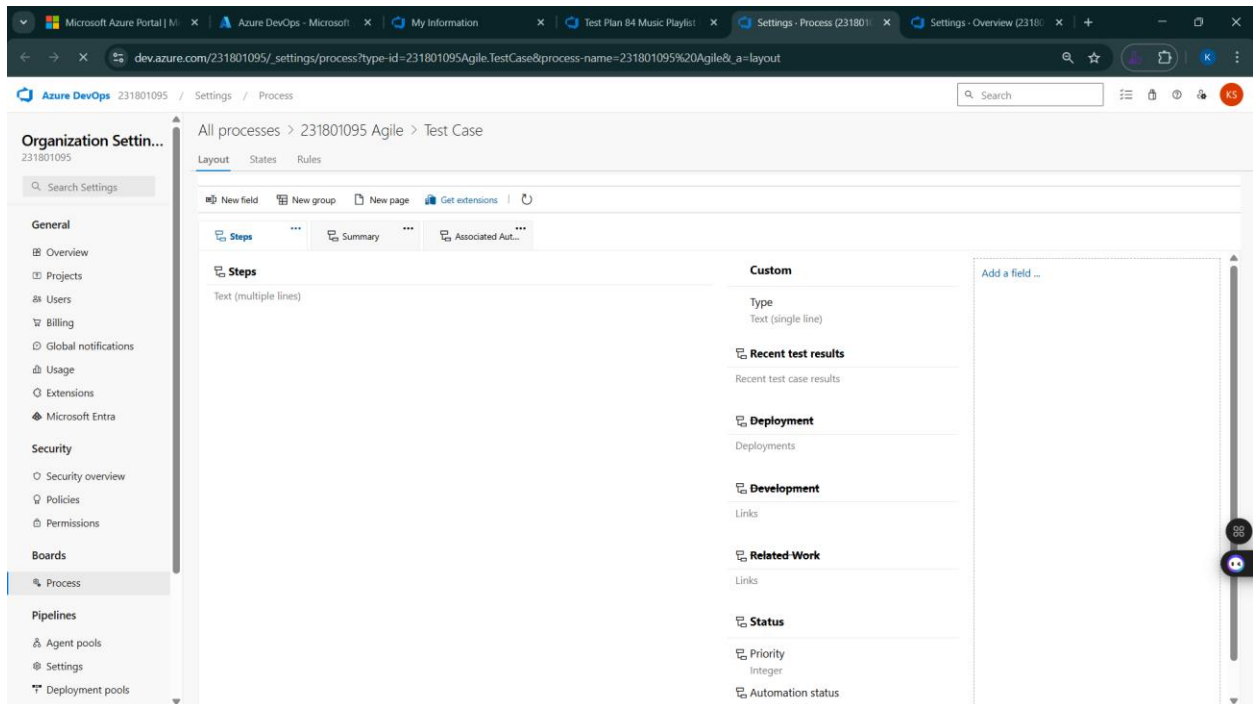
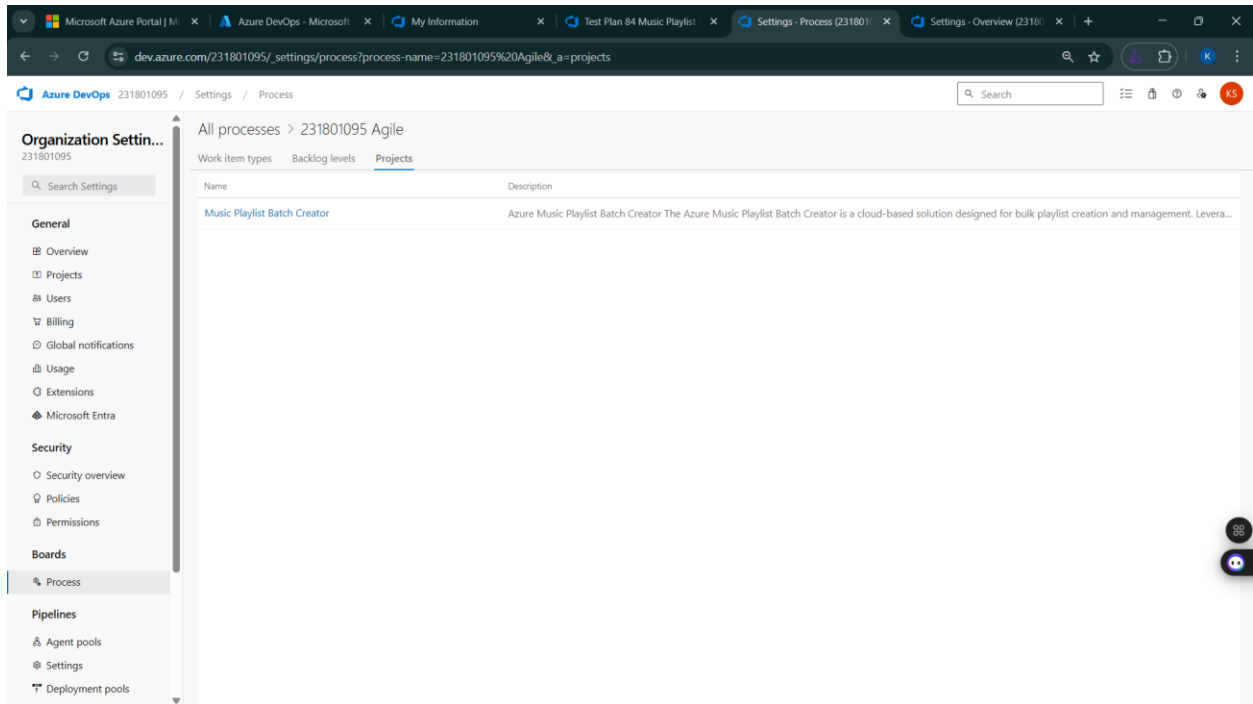
Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of process.	0



12.View the new test case template





Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9	LOAD TESTING AND PIPELINES
------------------	-----------------------------------

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.

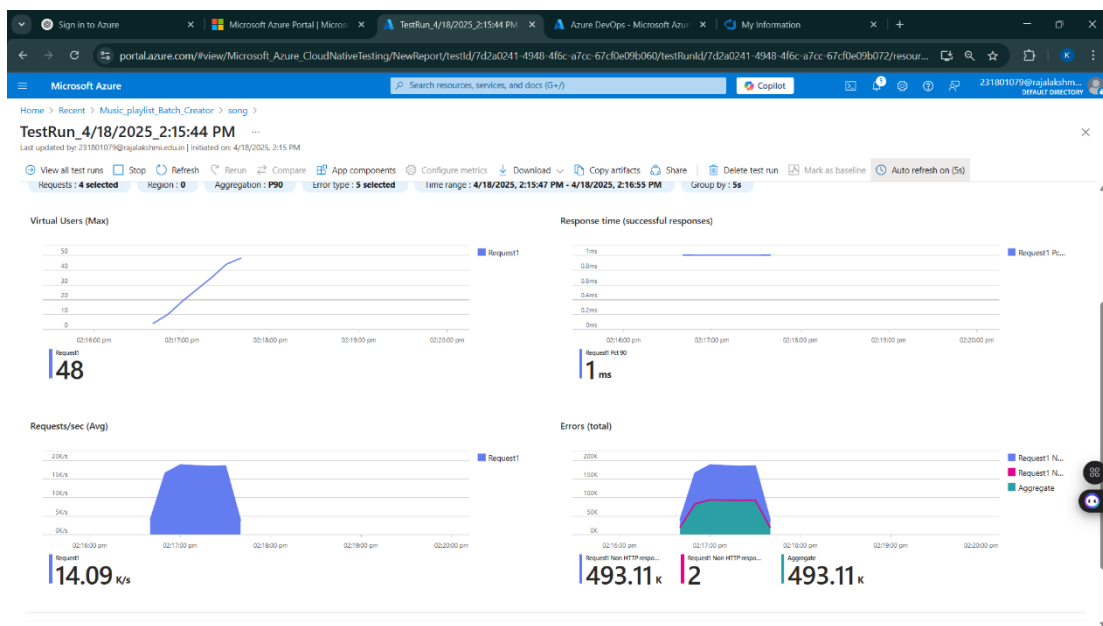
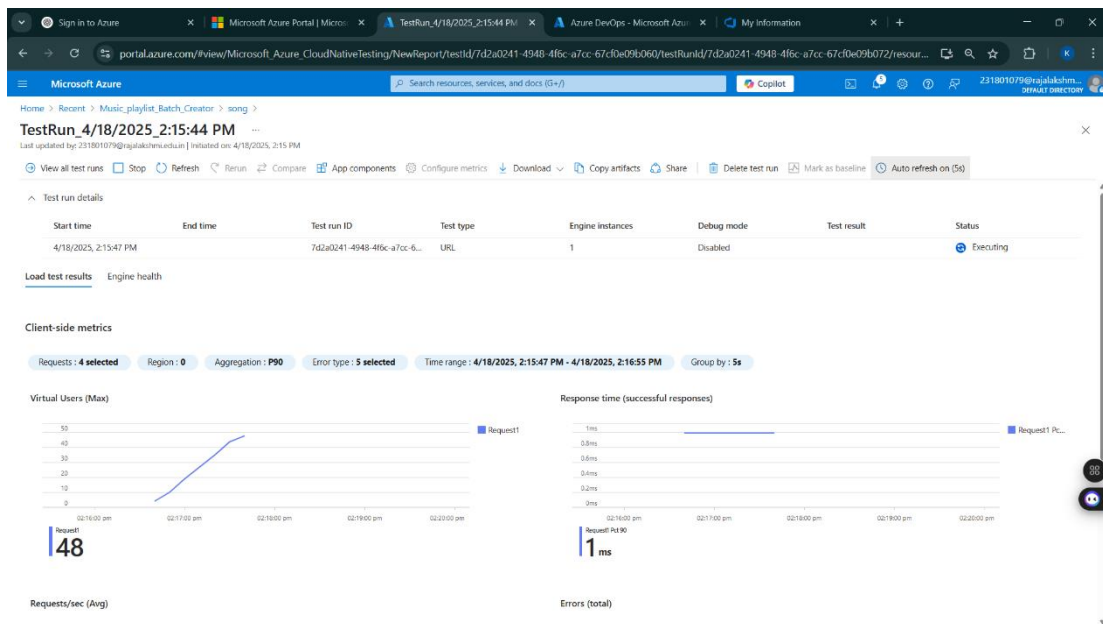
2. Create the Resource
 - Go to *Create a resource* → Search for “Azure Load Testing”.
 - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - *Subscription*: Choose your Azure subscription.
 - *Resource Group*: Create new or select an existing one.
 - *Name*: Provide a unique name (no special characters).
 - *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

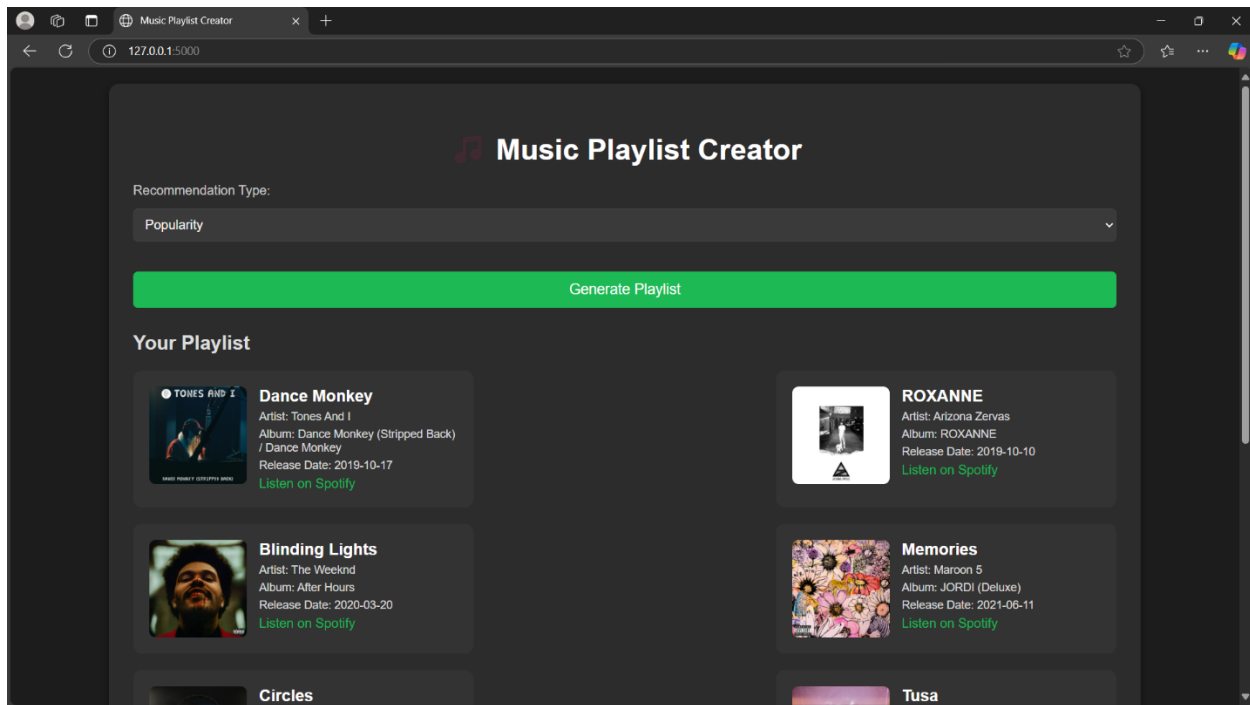
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - *Test Name*: Provide a unique name.
 - *Description*: (Optional) Add test purpose.
 - *Run After Creation*: Keep checked.
3. Load Settings
 - *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - In Azure DevOps, create a new project.
 - Create a pipeline and select GitHub as the source.
 - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - Add the following basic pipeline configuration for Python and Flask:

yml Code

trigger:

- main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

Step 1: Checkout the code from GitHub

- checkout: self

Step 2: Set up Python environment

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

Step 3: Install dependencies from the correct path

- script: |

python -m pip install --upgrade pip

pip install -r project/requirements.txt # Adjusted path to requirements.txt

displayName: "Install dependencies"

Step 4: Run a simple Python script to check the environment

- script: |

python -c "print('🎵 Hello from Music Playlist Batch Creator!)"

displayName: "Run a Python script"

3. Pipeline Tasks Include:

- Setting up the Python environment using the UsePythonVersion task.
- Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot displays the Azure DevOps web interface for a pipeline run. The browser address bar shows the URL: `dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_build/results?buildId=308&view=results`. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines (selected), Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The main content area shows the details for Pipeline 2, run #20250424.3. A message states: "This run is being retained as one of 3 recent runs by main (Branch)." with a "View retention leases" link. Below this, the "Summary" tab is active, showing it was manually run by Karthick S. The repository is "Music Playlist Batch Creator" at path "main" with commit "a87bd670". The run started "Just now" and took "24s". It has "0 work items" and "0 artifacts". A "View 52 changes" link is present. The "Jobs" section shows a single job named "Job" with a status of "Success" and a duration of "6s".

#20250424.3 • Pipeline 2

This run is being retained as one of 3 recent runs by main (Branch). [View retention leases](#)

Summary | Code Coverage

Manually run by **Karthick S** [View 52 changes](#)

Repository and version: **Music Playlist Batch Creator** **main** **a87bd670**

Time started and elapsed: **Just now** **24s**

Related: **0 work items** **0 artifacts**

Tests and coverage: [Get started](#)

Jobs

Name	Status	Duration
Job	Success	6s

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

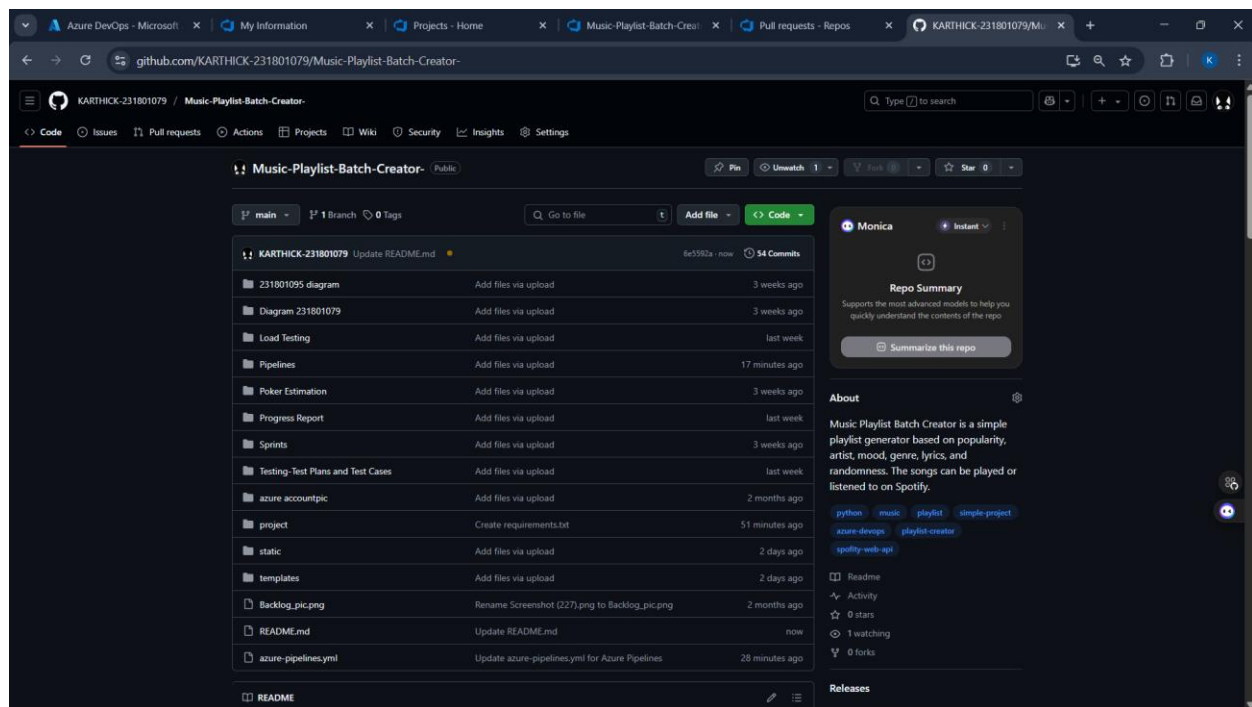
EXP NO: 10

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Music Playlist Batch Creator project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.